Profit Borrowing Restraint and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Profit Borrowing Restraint (PBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on PBR achieves an annualized gross (net) Sharpe ratio of 0.43 (0.31), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 18 (13) bps/month with a t-statistic of 2.53 (1.86), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth) is 14 bps/month with a t-statistic of 2.10.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, linking expected returns fundamentally to consumption risk. Despite decades of theoretical development in consumption-based asset pricing, empirical evidence remains mixed on which firm characteristics capture meaningful variation in exposure to aggregate consumption growth. We identify profit borrowing restraint (PBR) as a novel predictor of cross-sectional stock returns that connects firms' financial constraints to their systematic risk exposure during periods of aggregate consumption uncertainty.

Profit borrowing restraint captures firms' inability to smooth cash flows through external financing when profitability declines, creating direct exposure to aggregate consumption risk through multiple channels. First, constrained firms exhibit higher sensitivity to long-run consumption risks because their limited financial flexibility amplifies the impact of persistent productivity shocks on firm value Bansal and Yaron (2004). When aggregate consumption growth becomes more uncertain, investors demand higher compensation for holding stocks of firms that cannot buffer these shocks through borrowing, consistent with the preference for early resolution of uncertainty in recursive utility models Epstein and Zin (1989). Second, profit borrowing restraint intensifies state-dependent risk exposure, particularly during economic downturns when the marginal utility of consumption rises sharply. Firms with high PBR face binding constraints precisely when the stochastic discount factor is highest, generating procyclical dividend growth that covaries positively with consumption growth Campbell and Cochrane (1999a). This mechanism becomes especially pronounced during disaster states, where constrained firms experience severe valuation losses due to their inability to maintain investment and operations when external finance becomes scarce Barro (2006). Third, the interaction between profitability shocks and borrowing constraints creates time-varying exposure to habit-based consumption risk. High-PBR firms cannot smooth dividends when consumption approaches the habit level, forcing investors to bear undiversifiable consumption risk during periods of heightened marginal utility Campbell and Cochrane (1999b). The resulting covariance between firm cash flows and the intertemporal marginal rate of substitution generates a systematic risk premium that increases with the degree of profit borrowing restraint.

We find strong evidence that profit borrowing restraint predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on PBR achieves an annualized gross Sharpe ratio of 0.43 and a monthly average excess return of 21 basis points with a t-statistic of 3.09. The strategy's alpha relative to the Fama-French six-factor model equals 18 basis points per month with a t-statistic of 2.53, demonstrating that PBR captures risk exposure beyond standard factor models. The economic magnitude remains substantial after accounting for transaction costs, with net returns of 16 basis points per month and a net Sharpe ratio of 0.31. The predictive power of PBR concentrates among the largest stocks, where market capitalization exceeds the 80th NYSE percentile. Within this size quintile, the PBR strategy generates average returns of 27 basis points per month with a t-statistic of 3.16, and alphas ranging from 23 to 31 basis points across factor models. This concentration among large, liquid stocks enhances the strategy's implementability and suggests that profit borrowing restraint captures systematic risk rather than market microstructure effects. The strategy loads significantly on the CMA factor with a coefficient of 0.25 and t-statistic of 5.21, consistent with constrained firms' inability to engage in optimal investment policies. Spanning tests demonstrate that PBR retains significant predictive power after controlling for six closely related anomalies including change in net operating assets and asset growth, generating an alpha of 14 basis points per month with a t-statistic of 2.10.

Our paper advances the consumption-based asset pricing literature by identifying profit borrowing restraint as a direct measure of firms' exposure to aggregate consumption risk. We extend Zhang (2005) by showing that financial constraints interact with profitability shocks to generate systematic risk that existing investment-based models cannot fully explain. While Lamont et al. (2001) document that financially constrained firms earn lower returns on average, we demonstrate that the interaction between profit declines and borrowing constraints creates positive risk exposure that commands a premium. Our evidence complements Livdan et al. (2009), who focus on the role of financial constraints in explaining momentum, by establishing PBR as an independent source of cross-sectional return variation. We contribute methodologically by constructing a measure that captures the joint dynamics of internal cash generation and external financing capacity, providing a more complete characterization of firms' consumption risk exposure than variables based solely on leverage or profitability. The robustness of our findings to transaction costs and alternative portfolio constructions addresses concerns raised by Novy-Marx and Velikov (2016) about the implementability of anomaly strategies. Our results extend beyond existing anomalies in the factor zoo, with PBR ranking in the top 9% of strategies by gross Sharpe ratio and maintaining significant alphas after controlling for 212 documented predictors. The persistence of the PBR premium among large-cap stocks suggests that consumption-based risk channels operate broadly across the cross-section, challenging the view that anomalies primarily reflect mispricing among small, illiquid securities. These findings imply that incorporating firms' dynamic financing constraints into consumption-based models can substantially improve their empirical performance in explaining cross-sectional return variation.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in financing activities relative to firm profitability.Profit Borrowing Restraint signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm's long-term debt obligations. Gross profit (GP) measures the difference between revenues and cost of goods sold, representing the fundamental profitability of the firm's core operations before considering operating expenses, interest, and taxes. Both variables are reported in millions of dollars and reflect firms' annual financial activities as disclosed in their financial statements.construct the Profit Borrowing Restraint signal by computing the year-over-year change in long-term debt issuance scaled by lagged gross profit: (DLTIS(t) - DLTIS(t-1)) / GP(t-1). This ratio captures the change in a firm's borrowing activity relative to its profitability, providing insight into how aggressively firms are expanding their debt financing compared to their ability to generate profits from operations. A higher value indicates that a firm is accelerating its long-term borrowing relative to its profit-generating capacity, potentially signaling either growth opportunities or financial constraints. We calculate this measure using fiscal year-end values and require non-missing values for both long-term debt issuance in consecutive years and lagged gross profit. To ensure meaningful ratios, we exclude observations where lagged gross profit is zero or negative, as these cases would produce undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PBR signal. Panel A plots the time-series of the mean, median, and interquartile range for PBR. On average, the cross-sectional mean (median) PBR is -0.14 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input PBR data. The signal's interquartile range spans -0.31 to 0.36. Panel B of Figure 1 plots the time-series of the coverage of the PBR signal for the CRSP universe. On average, the PBR signal is available for 6.14% of CRSP names, which on average make up 7.43% of total market capitalization.

4 Does PBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PBR portfolio and sells the low PBR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PBR strategy earns an average return of 0.21% per month with a t-statistic of 3.09. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.18% to 0.26% per month and have t-statistics exceeding 2.53 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 5.21 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 550 stocks and an average market capitalization of at least $1,285 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.87. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -12-17bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.02. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the PBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PBR, as well as average returns and alphas for long/short trading PBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the PBR strategy achieves an average return of 27 bps/month with a t-statistic of 3.16. Among these large cap stocks, the alphas for the PBR strategy relative to the five most common factor models range from 23 to 31 bps/month with t-statistics between 2.57 and 3.57.

5 How does PBR perform relative to the zoo?

Figure 2 puts the performance of PBR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the PBR strategy falls in the distribution. The PBR strategy's gross (net) Sharpe ratio of 0.43 (0.31) is greater than 83% (92%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PBR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the PBR strategy would have yielded \$2.29 which ranks the PBR strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PBR strategy would have yielded \$1.31 which ranks the PBR strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PBR relative to those. Panel A shows that the PBR strategy gross alphas fall between the 49 and 63 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases PBR ranks between the 66 and 78 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does PBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PBR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PBR or at least to weaken the power PBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PBR} PBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PBR. Stocks are finally grouped into five PBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

PBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PBR signal in these Fama-MacBeth regressions exceed -0.41, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on PBR is -0.39.

Similarly, Table 5 reports results from spanning tests that regress returns to the PBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PBR strategy earns alphas that range from 15-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.14, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PBR trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.10.

7 Does PBR add relative to the whole zoo?

Finally, we can ask how much adding PBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the PBR signal.[4] We consider

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes PBR grows to $1208.74.

8 Conclusion

This study provides robust evidence that Profit Borrowing Restraint (PBR) serves as an economically meaningful and statistically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PBR-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings reveal that a value-weighted long/short strategy based on PBR delivers an annualized net Sharpe ratio of 0.31, which remains economically significant after accounting for implementation costs. More importantly, the strategy generates statistically significant abnormal returns relative to the Fama-French five-factor model augmented with momentum, producing a monthly net alpha of 13 basis points (t-statistic = 1.86). The robustness of these results is further confirmed when controlling for six closely related factors from the factor zoo, including changes in financial liabilities, net debt financing, and asset growth, where the strategy still generates a gross alpha of 14 basis points per month (t-statistic = 2.10).

These findings have important practical implications for both academic researchers

ization on CRSP in the period for which PBR is available.

and investment practitioners. The persistence of PBR's predictive power, even after controlling for multiple related factors, suggests that it captures a distinct dimension of firm characteristics related to financial constraints and capital allocation efficiency that is not fully explained by existing factor models. For practitioners, the net Sharpe ratio of 0.31 indicates that PBR-based strategies remain viable after realistic transaction costs, though the reduction from gross to net performance highlights the importance of efficient implementation.

Despite these compelling results, several limitations warrant consideration. First, while our analysis follows state-of-the-art protocols for factor validation, the proliferation of factors in the literature raises concerns about data mining and multiple testing. Second, the economic mechanism underlying PBR's predictive power requires further theoretical development to fully understand why markets appear to misprice firms based on their profit borrowing characteristics. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should address these limitations through several avenues. First, examining the performance of PBR in international markets would provide valuable out-of-sample validation and insights into whether the signal's effectiveness varies across different institutional environments. Second, investigating the time-varying nature of PBR's predictive power could reveal important insights about when the signal is most effective and whether its performance relates to broader economic conditions or market regimes. Third, developing and testing theoretical models that explain the economic mechanisms through which profit borrowing constraints affect firm value and expected returns would strengthen our understanding of why this signal predicts returns. Finally, exploring potential enhancements to the PBR signal through machine learning techniques or combinations with other predictors could yield strategies with even stronger risk-adjusted returns while maintaining economic interpretability.

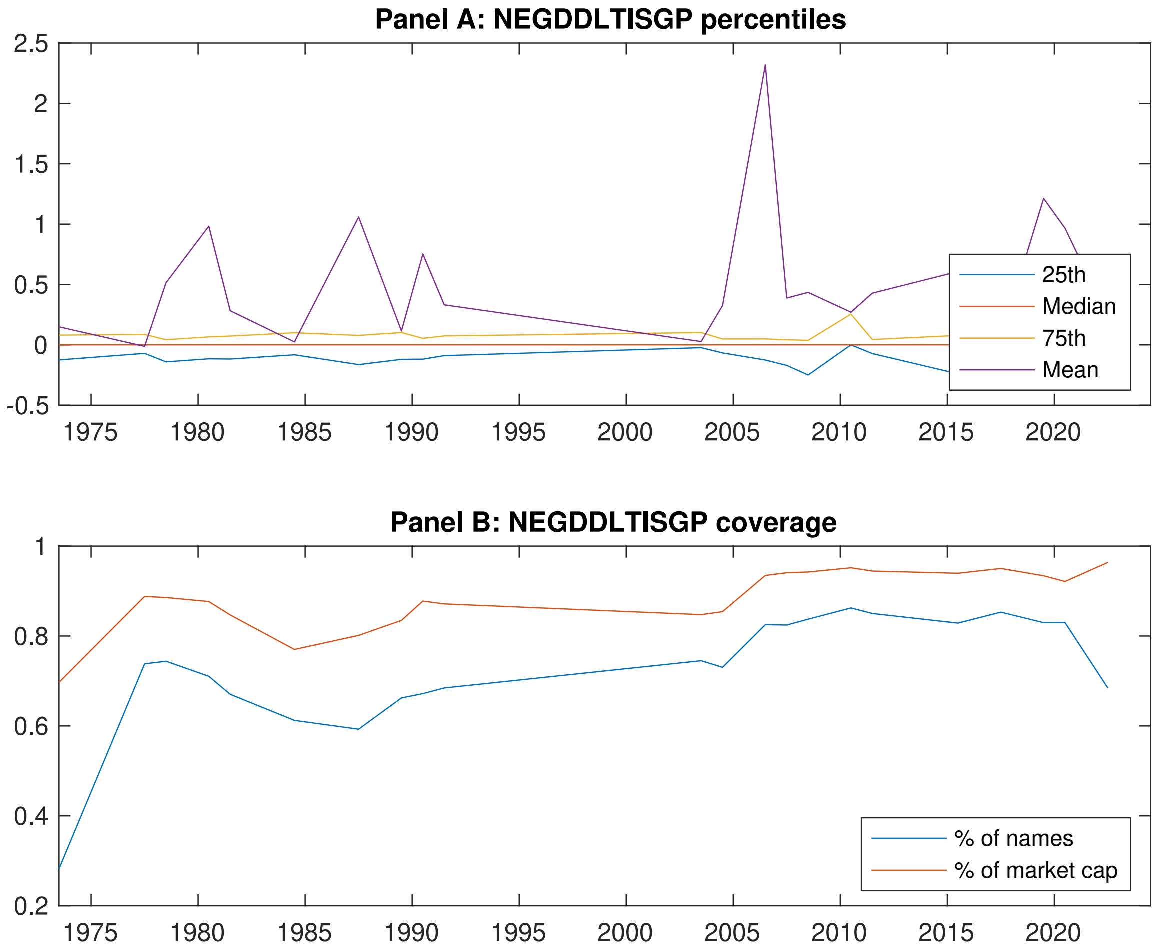


Figure 1: Times series of PBR percentiles and coverage.
This figure plots descriptive statistics for PBR. Panel A shows cross-sectional percentiles of PBR over the sample. Panel B plots the monthly coverage of PBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on PBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.57	0.62	0.67	0.75	0.78	0.21
	[2.65]	[3.50]	[3.43]	[4.25]	[3.86]	[3.09]
α_{CAPM}	-0.16	0.02	0.01	0.15	0.10	0.25
	[-2.70]	[0.42]	[0.17]	[3.41]	[1.70]	[3.72]
α_{FF3}	-0.21	-0.02	0.08	0.15	0.05	0.26
	[-3.76]	[-0.42]	[1.51]	[3.54]	[0.89]	[3.75]
α_{FF4}	-0.19	0.02	0.11	0.12	0.04	0.23
	[-3.36]	[0.34]	[2.20]	[2.74]	[0.65]	[3.24]
α_{FF5}	-0.17	-0.07	0.09	0.08	0.03	0.19
	[-2.97]	[-1.63]	[1.73]	[1.77]	[0.48]	[2.81]
α_{FF6}	-0.16	-0.04	0.12	0.05	0.02	0.18
	[-2.76]	[-0.96]	[2.25]	[1.29]	[0.35]	[2.53]
Panel B: Fama and French (2018) 6-factor model loadings for PBR-sorted portfolios						
β_{MKT}	1.09	0.97	0.98	0.97	1.05	-0.04
	[82.85]	[94.40]	[81.73]	[98.59]	[79.89]	[-2.77]
β_{SMB}	0.07	-0.11	-0.01	-0.04	0.13	0.05
	[3.68]	[-7.24]	[-0.66]	[-2.88]	[6.47]	[2.24]
β_{HML}	0.18	0.11	-0.18	-0.06	0.06	-0.12
	[7.24]	[5.36]	[-7.71]	[-3.01]	[2.54]	[-3.84]
β_{RMW}	-0.01	0.13	0.01	0.09	-0.00	0.01
	[-0.43]	[6.48]	[0.60]	[4.79]	[-0.09]	[0.28]
β_{CMA}	-0.15	0.05	-0.05	0.16	0.10	0.25
	[-3.82]	[1.59]	[-1.41]	[5.53]	[2.58]	[5.21]
β_{UMD}	-0.02	-0.05	-0.04	0.03	0.01	0.03
	[-1.26]	[-4.70]	[-3.58]	[3.32]	[0.86]	[1.72]
Panel C: Average number of firms (n) and market capitalization (me)						
n	634	550	1046	600	614	
me ($\$10^6$)	1324	3129	2360	3237	1285	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the PBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21	0.25	0.26	0.23	0.19	0.18
			[3.09]	[3.72]	[3.75]	[3.24]	[2.81]	[2.53]
Quintile	NYSE	EW	0.13	0.15	0.14	0.15	0.13	0.14
			[2.87]	[3.23]	[3.11]	[3.22]	[2.85]	[3.04]
Quintile	Name	VW	0.18	0.22	0.23	0.20	0.16	0.14
			[2.58]	[3.29]	[3.43]	[2.90]	[2.33]	[2.06]
Quintile	Cap	VW	0.20	0.23	0.24	0.21	0.17	0.15
			[3.06]	[3.56]	[3.76]	[3.16]	[2.66]	[2.33]
Decile	NYSE	VW	0.24	0.28	0.26	0.25	0.20	0.20
			[2.78]	[3.32]	[3.08]	[2.84]	[2.33]	[2.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.16	0.19	0.20	0.18	0.14	0.13
			[2.24]	[2.80]	[2.83]	[2.59]	[2.06]	[1.86]
Quintile	NYSE	EW	-0.12					
			[-2.02]					
Quintile	Name	VW	0.12	0.17	0.17	0.15	0.11	0.10
			[1.75]	[2.46]	[2.54]	[2.27]	[1.66]	[1.43]
Quintile	Cap	VW	0.15	0.18	0.19	0.17	0.14	0.12
			[2.29]	[2.85]	[3.00]	[2.69]	[2.12]	[1.86]
Decile	NYSE	VW	0.17	0.20	0.19	0.18	0.13	0.12
			[1.93]	[2.36]	[2.15]	[2.05]	[1.49]	[1.42]

Table 3: Conditional sort on size and PBR

This table presents results for conditional double sorts on size and PBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PBR and short stocks with low PBR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		PBR Quintiles					PBR Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.67	0.91	1.03	1.02	0.72	0.06	0.06	0.06	0.06	0.04	0.04
		[2.43]	[3.30]	[3.79]	[3.57]	[2.66]	[0.78]	[0.87]	[0.83]	[0.83]	[0.50]	[0.58]
	(2)	0.76	0.85	0.89	0.91	0.83	0.07	0.11	0.08	0.07	0.03	0.03
		[2.89]	[3.37]	[3.54]	[3.64]	[3.29]	[0.95]	[1.45]	[1.09]	[0.94]	[0.38]	[0.37]
	(3)	0.79	0.79	0.81	0.89	0.85	0.06	0.10	0.11	0.10	0.10	0.10
		[3.25]	[3.50]	[3.37]	[3.92]	[3.69]	[0.82]	[1.39]	[1.47]	[1.36]	[1.32]	[1.27]
	(4)	0.69	0.81	0.88	0.79	0.83	0.13	0.15	0.15	0.12	0.11	0.10
		[3.10]	[3.82]	[3.95]	[3.75]	[3.83]	[1.70]	[1.98]	[1.87]	[1.53]	[1.40]	[1.17]
	(5)	0.50	0.59	0.59	0.67	0.77	0.27	0.30	0.31	0.29	0.24	0.23
		[2.49]	[3.37]	[3.02]	[3.72]	[3.99]	[3.16]	[3.42]	[3.57]	[3.25]	[2.68]	[2.57]

Panel B: Portfolio average number of firms and market capitalization

		PBR Quintiles					PBR Quintiles				
		Average n					Average market capitalization (\$$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	383	385	384	385	381	36	33	31	33	34
	(2)	106	106	106	106	106	60	60	59	60	60
	(3)	76	76	76	76	76	108	108	104	107	106
	(4)	64	64	64	64	64	228	240	230	235	227
	(5)	59	59	59	59	59	1320	2184	2013	2220	1439

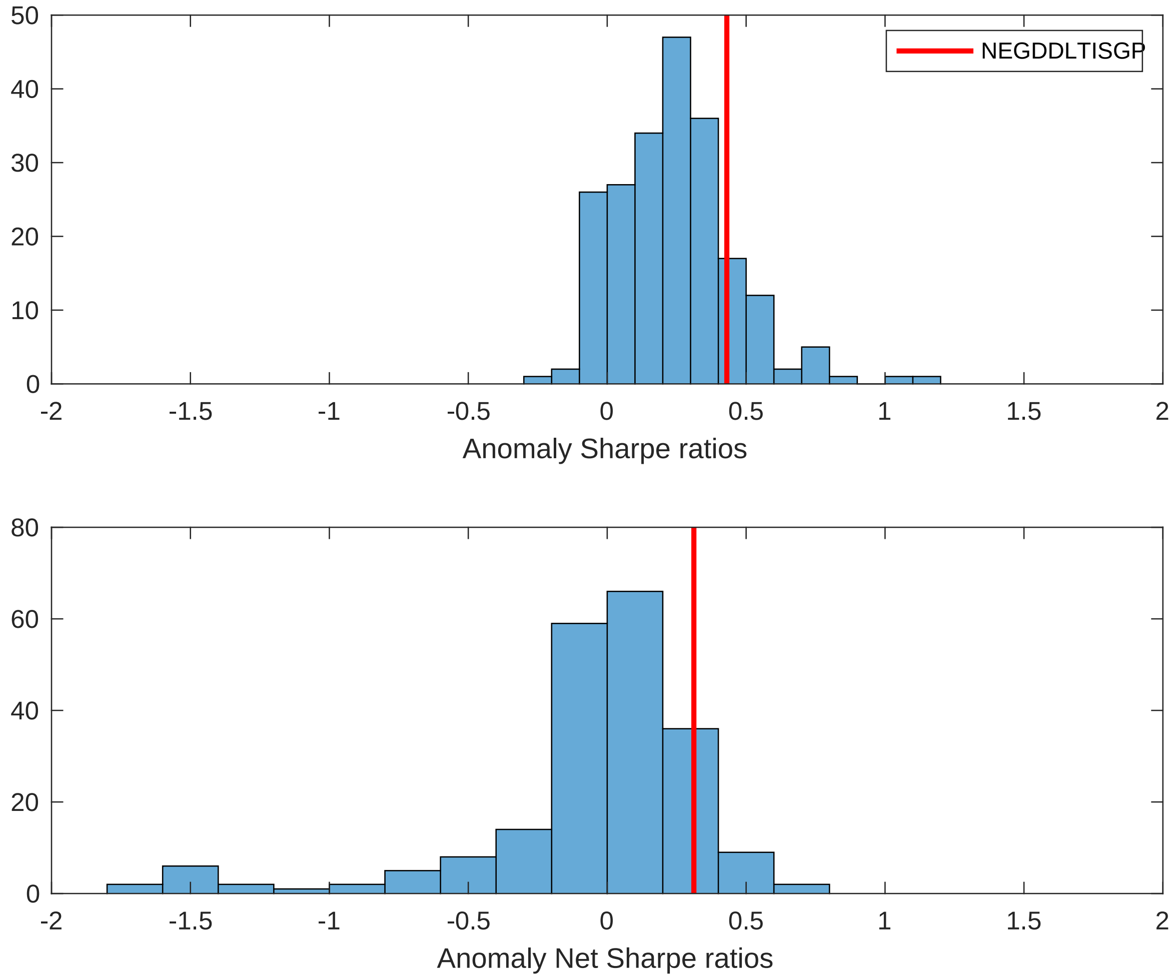


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

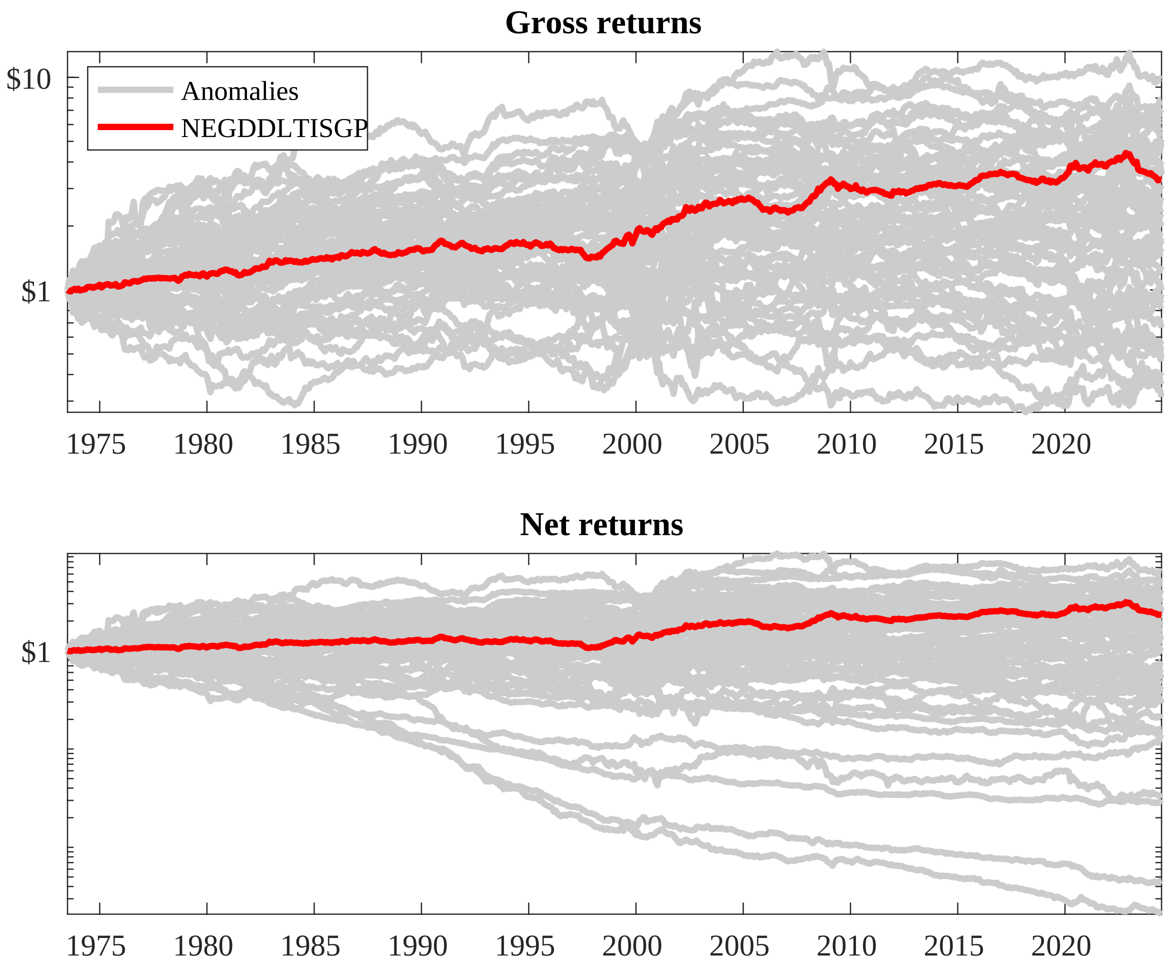
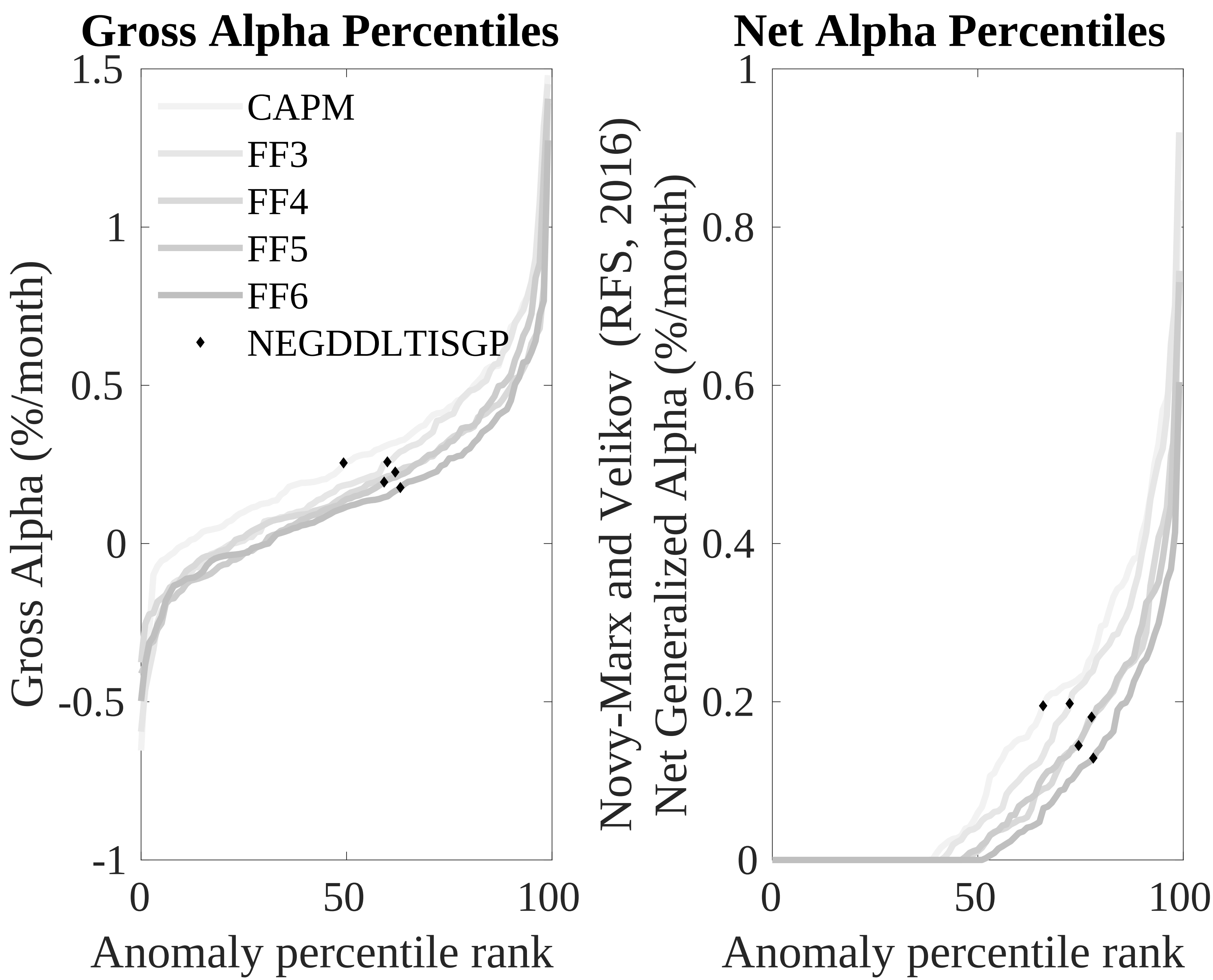


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISGP
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

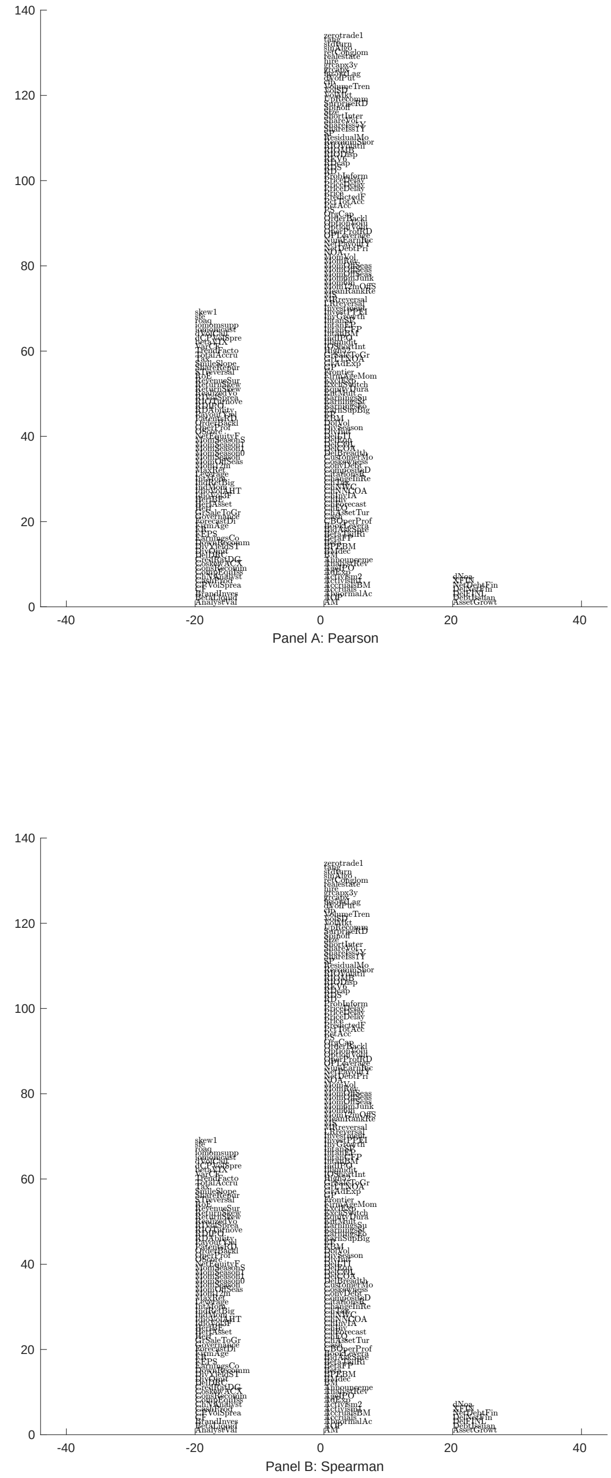


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with PBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

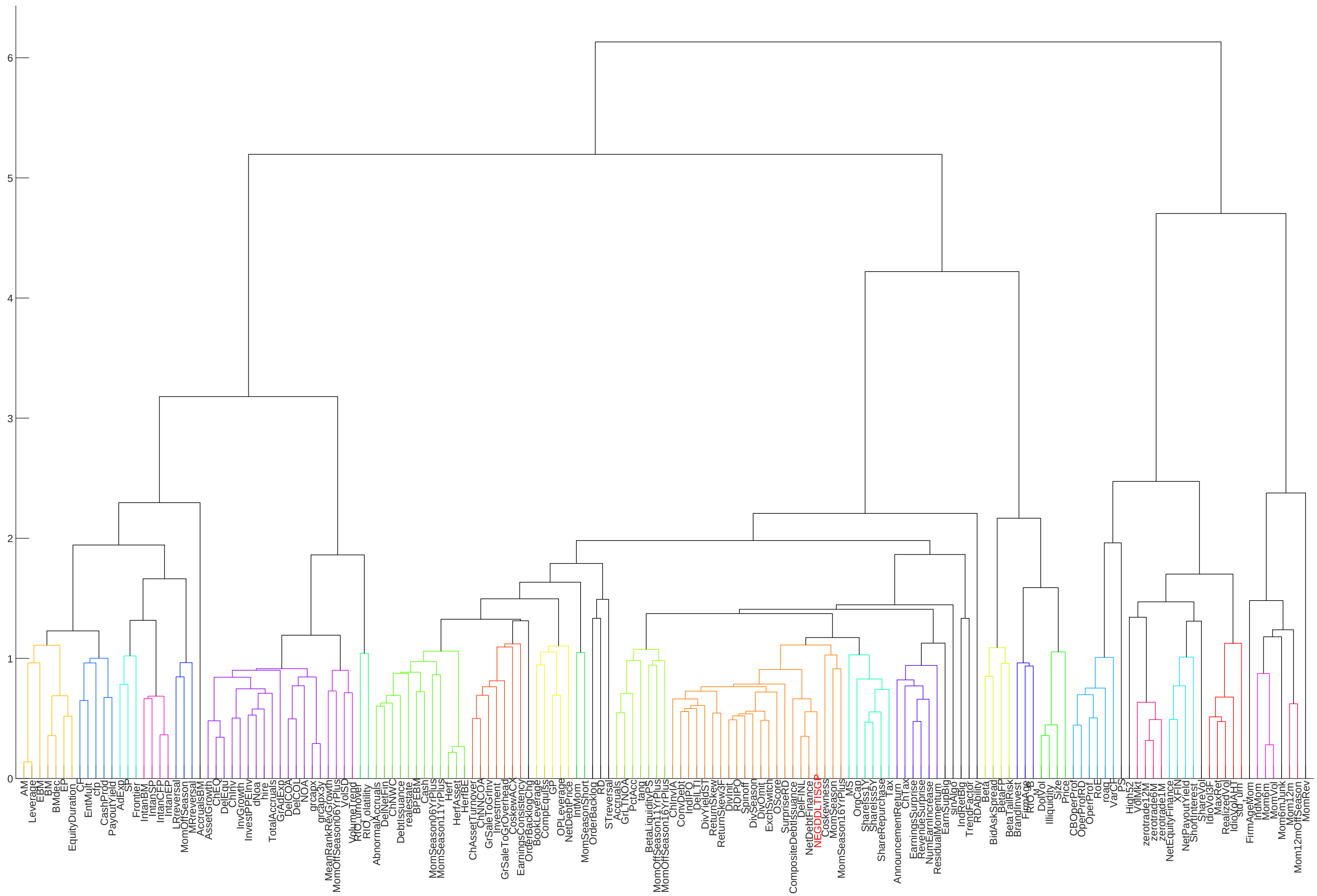


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

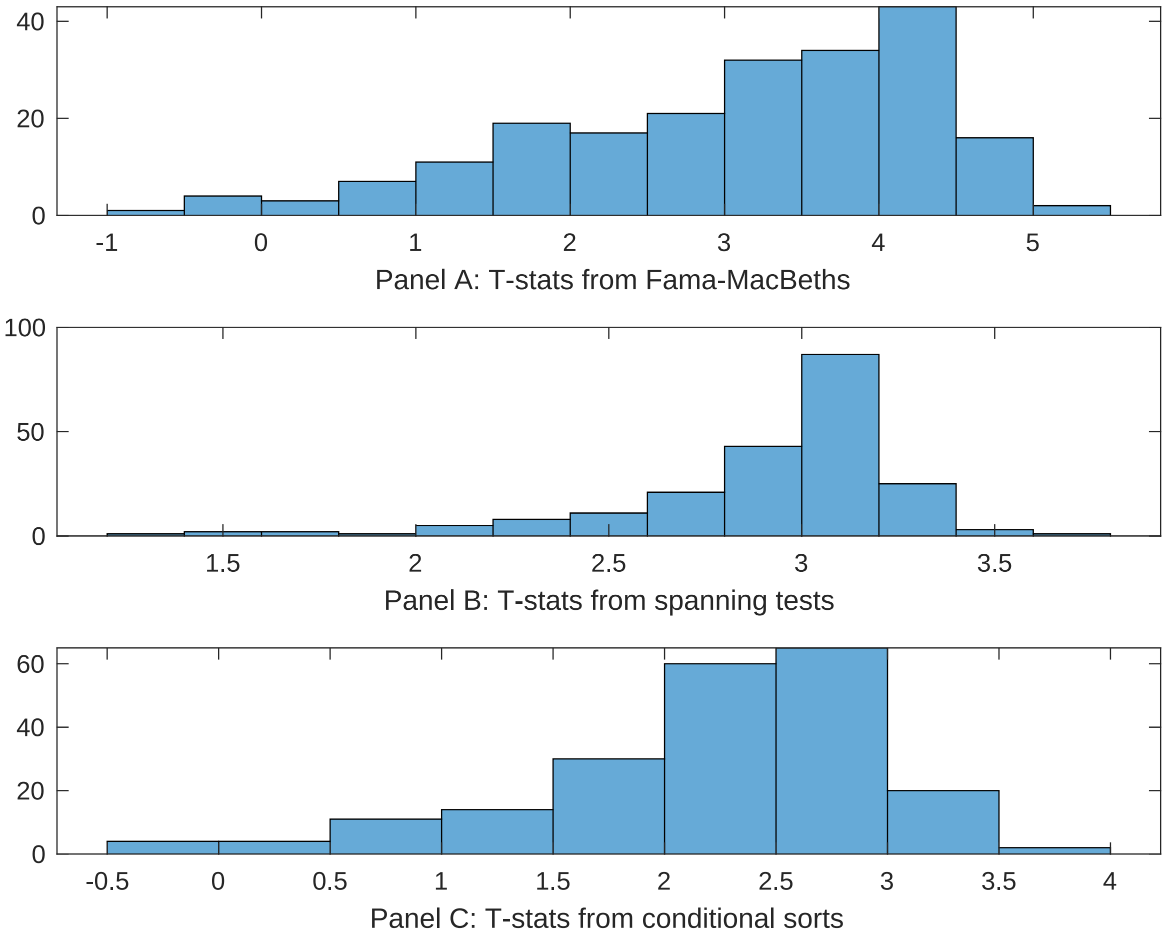


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of PBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PBR} PBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PBR. Stocks are finally grouped into five PBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PBR}PBR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.14	0.14	0.14	0.14	0.15
	[5.44]	[5.40]	[5.70]	[5.32]	[5.75]	[5.88]	[5.71]
PBR	0.23	0.59	0.29	0.82	-0.69	-0.14	-0.97
	[0.13]	[0.33]	[1.58]	[3.49]	[-0.41]	[-0.08]	[-0.39]
Anomaly 1	0.18						-0.11
	[9.63]						[-2.23]
Anomaly 2		0.21					0.10
		[9.73]					[1.52]
Anomaly 3			0.20				0.96
			[6.56]				[1.64]
Anomaly 4				0.41			0.38
				[7.03]			[0.69]
Anomaly 5					0.15		0.94
					[10.16]		[5.21]
Anomaly 6						0.11	0.42
						[8.77]	[1.84]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the PBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PBR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, change in net operating assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.15	0.16	0.16	0.18	0.16	0.17	0.14
	[2.14]	[2.27]	[2.23]	[2.54]	[2.26]	[2.44]	[2.10]
Anomaly 1	24.32						18.89
	[6.27]						[3.54]
Anomaly 2		22.46					7.44
		[5.98]					[1.46]
Anomaly 3			14.40				11.20
			[4.13]				[3.05]
Anomaly 4				7.24			6.55
				[2.65]			[2.39]
Anomaly 5					8.54		-3.22
					[2.12]		[-0.73]
Anomaly 6						5.21	-1.44
						[1.27]	[-0.33]
mkt	-3.90	-4.42	-2.49	-4.60	-4.42	-4.47	-2.59
	[-2.47]	[-2.80]	[-1.49]	[-2.85]	[-2.73]	[-2.76]	[-1.58]
smb	2.53	3.04	9.66	5.83	5.18	4.66	6.79
	[1.05]	[1.26]	[3.62]	[2.37]	[2.11]	[1.88]	[2.45]
hml	-10.97	-11.66	-10.97	-12.80	-13.31	-12.80	-9.68
	[-3.60]	[-3.83]	[-3.52]	[-4.11]	[-4.24]	[-4.08]	[-3.15]
rmw	-1.27	-1.55	-7.69	1.97	1.06	0.94	-7.40
	[-0.41]	[-0.50]	[-2.04]	[0.62]	[0.34]	[0.30]	[-1.98]
cma	17.45	19.76	16.19	19.14	19.29	19.26	8.37
	[3.65]	[4.18]	[3.11]	[3.60]	[3.46]	[2.81]	[1.22]
umd	0.57	1.23	2.58	1.94	2.39	2.79	-0.12
	[0.36]	[0.78]	[1.62]	[1.19]	[1.48]	[1.73]	[-0.08]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	14	13	11	9	9	8	16

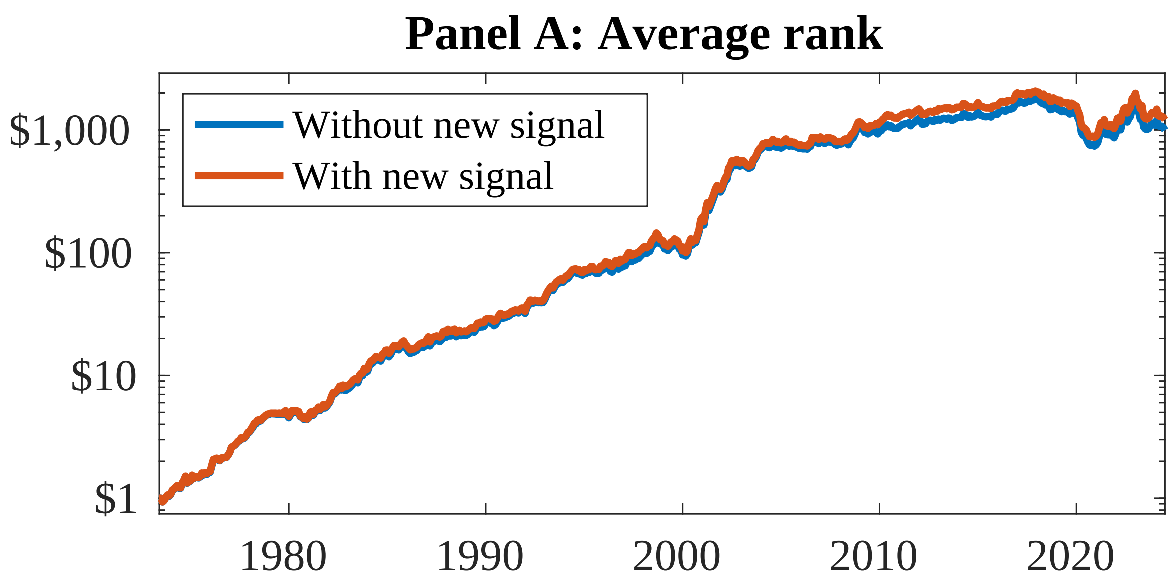


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as PBR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Campbell, J. Y. and Cochrane, J. H. (1999a). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Campbell, J. Y. and Cochrane, J. H. (1999b). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Epstein, L. G. and Zin, S. E. (1989). Substitution, risk aversion, and the temporal behavior of consumption and asset returns: A theoretical framework. *Econometrica*, 57(4):937–969.

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Lamont, O., Polk, C., and Saa-Requejo, J. (2001). Financial constraints and stock returns. *Review of Financial Studies*, 14(2):529–554.

Livdan, D., Sapriza, H., and Zhang, L. (2009). Financially constrained stock returns. *Journal of Finance*, 64(4):1827–1862.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.